

Program: Diploma in Fire Technology and Safety	
Course Code: 3468	Course Title: Advanced CADD Lab
Semester: 3	Credits: Nil
Course Category: Program Core	
Periods per week: 4 (L:0 T:0 P:4)	Periods per semester: 60

Course Objectives:

- To introduce CAD tools used in fire and safety system design.
- To apply Building Information Modeling (BIM) for designing fire protection systems like sprinklers.
- To model and visualize plant equipment, piping, and utilities related to fire safety.

Course Prerequisites:

Topic	Course code	Coursename	Semester
Basic computer knowledge		Introduction to IT systems	S1
Knowledge of Engineering Graphics		Engineering Graphics	S1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Use CAD software for designing fire safety systems.	13	Applying
CO2	Design sprinkler systems using BIM software for buildings.	14	Applying
CO3	Model piping, utilities, and fire protection equipment using CAD.	15	Applying
CO4	Simulate fire safety systems and assess their effectiveness.	14	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	1	1	2	2
CO2	3	3	2	2	2	2	3
CO3	2	3	3	2	3	2	2
CO4	2	2	3	3	2	3	1

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	Learn basic CAD techniques for fire and safety system designs		
M1.01	Basic drawing, editing, and 2D geometry using CAD.	4	Applying
M1.02	Create isometric and orthographic projections for fire safety systems.	5	Applying
M1.03	Part modeling and dimensioning for fire safety components.	4	Applying
CO2	Design fire protection systems using BIM software		
M2.01	Design sprinkler systems for buildings using BIM.	7	Understanding
M2.02	Apply industry standards to fire safety designs.	7	Understanding
	LabTestI	2	
CO3	Model fire protection equipment and piping systems		
M3.01	Basics of P&ID and fire safety systems.	5	Understanding
M3.02	Model 3D fire protection facilities and utilities.	5	Applying
M3.03	Pipe routing and visualization in CAD.	5	Applying

CO4	Simulate fire safety systems and assess performance		
M4.01	Use open-source software for fire modeling (zone & field models).	8	Analyzing
M4.02	Analyze fire simulation data for system effectiveness.	6	Analyzing
	LabTest II	2	

Text / Reference:

T/R	Book Title / Author
T1	AutoCAD Essentials, Autodesk official Press, John Wiley & Sons, USA
R2	Computer Aided Design Laboratory by M.N. Sessa Prakash & Dr. G.S. Servesesh – Laxmi Publications.
R3	Engineering Graphics by P.J. Sha–S. Chand & Co.
R4	Introduction to AutoCAD 2018 for Civil Engineering Applications by Nighat Yasmin, SDC Publications

Online Resources:

Sl.No	Website Link
1	https://www.autodesk.in/campaigns/autocad-tutorials
2	https://thesourcecad.com/autocad-tutorials/
3	https://all3dp.com/1/autocad-tutorial-for-beginners/
4	https://civilengineerspk.com/autocad-video-tutorials/